

Tank Level Control using PLC

Key Objectives

- Develop a ladder logic program using PLCnext to automate a tank filling and discharge process.
- Integrate analog sensors to detect tank levels and control valves accordingly.
- Simulate the process using Factory IO with visual indicators.
- Implement emergency stop (S3), manual discharge (S2), and LED status indicators.
- Display tank level numerically and manage valve control with precise real number conversions.

Technical Overview

The system was developed in PLCnext Engineering and integrated with Factory IO. The analog tank was configured to fill up to 300 cm when the Start button (S1) is pressed. Once full, the tank holds momentarily (LED4 ON), then discharges to 150 cm, holds again, and finally empties. Valves are controlled via real-value multipliers to match Factory IO's 0–10 range. LED1 flashes to indicate filling and LED3 for discharging. A dedicated emergency stop (S3) halts all activity, and S2 forces a full drain. Visual feedback is shown through Factory IO's lamps and level meter.

Highlights

- Real-world PLCnext controller (2152 Rev.0.0) and IO modules used.
- Accurate simulation with real-to-integer conversions to reflect true tank height.
- Emergency stop implementation reliably resets latches, timers, and LEDs.
- Responsive control logic allows manual override without disrupting sequence.
- Use of Factory IO digital indicators and analog level scaling (x30) to reflect tank height (0–300 cm).

Visual Aids

On the next page.

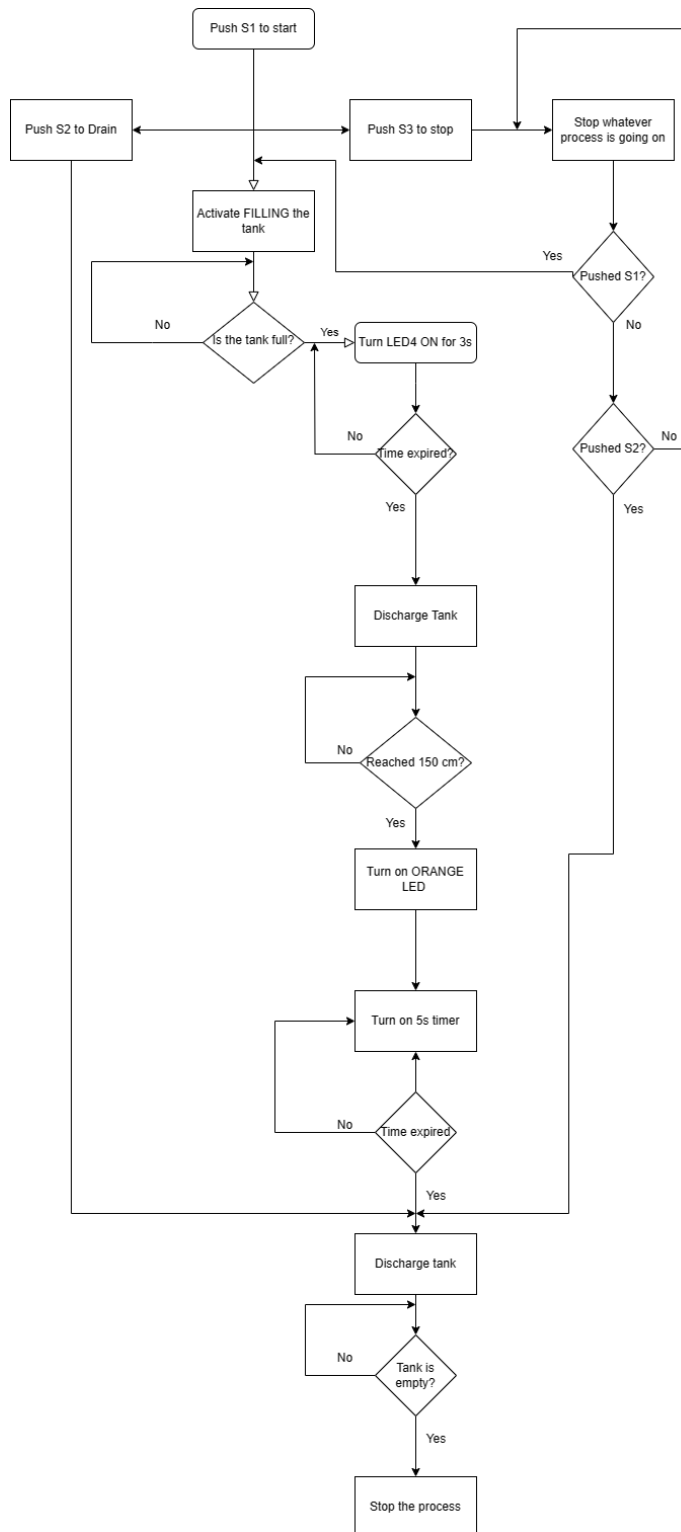
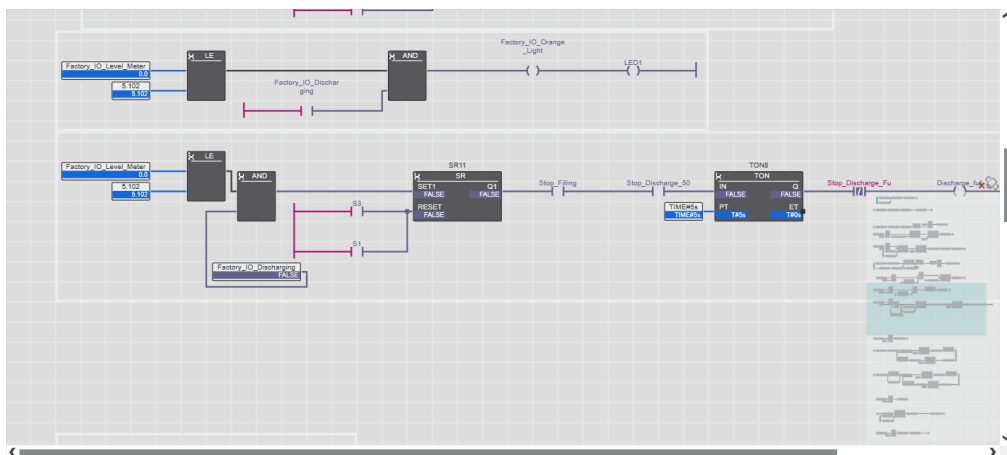
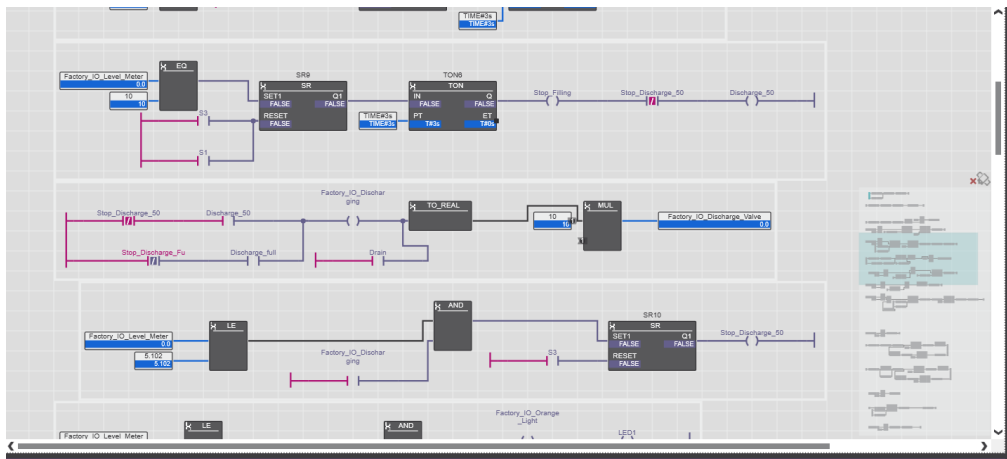
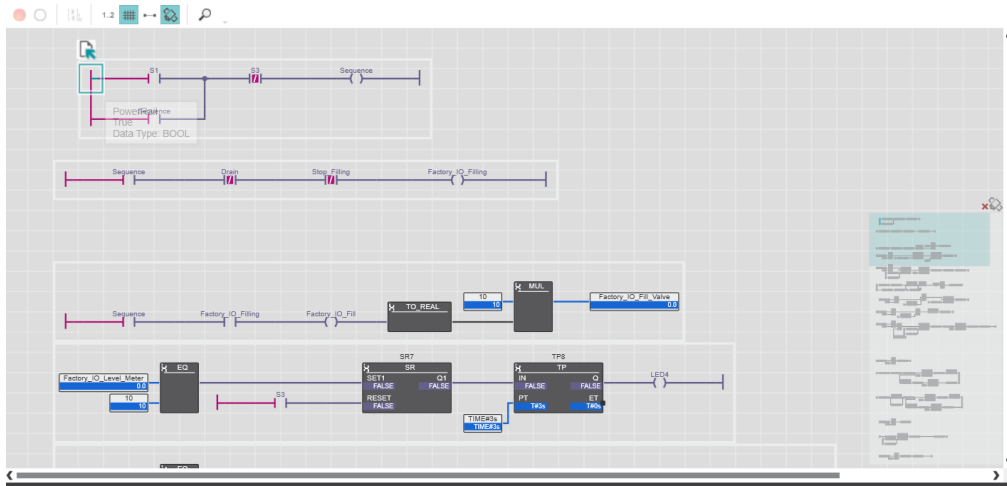


Figure 1: Flowchart for the code

Code



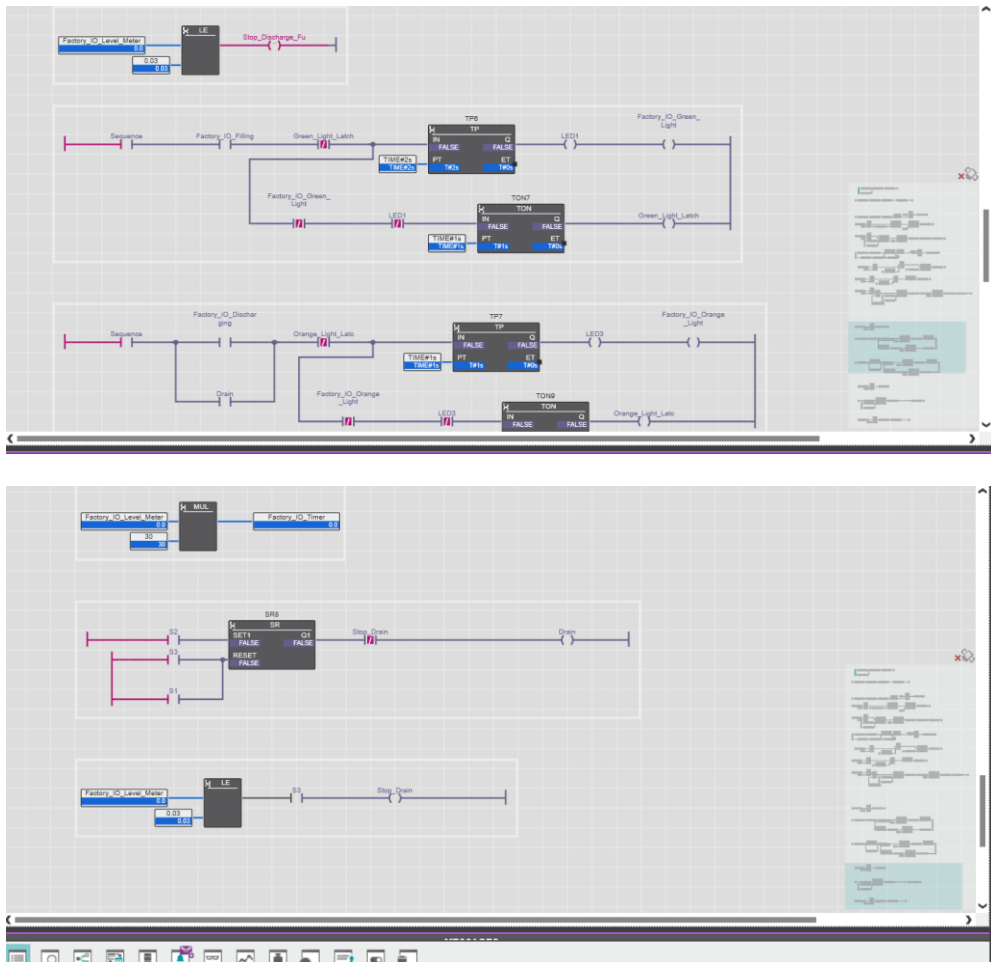


Figure 2: Full ladder logic code divided into five screenshots

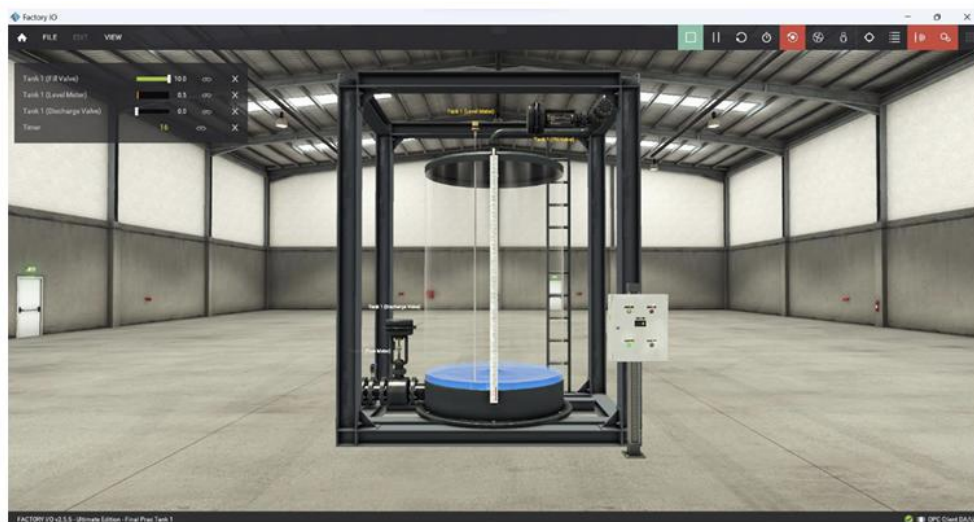


Figure 3: Tank simulation using Factory IO simulation environment

Conclusion

The system successfully met most design criteria. Automated filling and staged discharge function as expected, with synchronized visual feedback. Manual drain and emergency stop were effectively implemented. One minor issue was the ORANGE lamp not lighting during forced drain using S2. Despite this, the program demonstrated strong understanding of PLC logic, analog control, and simulation integration with Factory IO.